

High-speed Gaze-oriented Projection by Cross-ratio-based Eye Tracking with Dual Infrared Imaging

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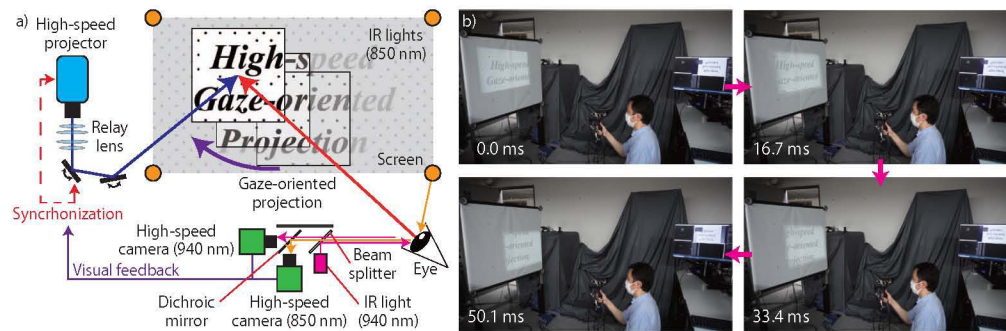


Figure 1: The proposed system; a) system configuration, b) demonstration of high-speed gaze-oriented projection.

ABSTRACT

While gaze-oriented projection can be high-resolution and wide-area display, conventional methods have difficulties in handling quick human eye movements. In this paper, we propose a high-speed gaze-oriented projection system using a synchronized high-speed tracking projector and cross-ratio-based eye tracking. The tracking projector with a high-speed projector and rotational mirrors enables temporal geometric consistency of the projection. The eye tracking uses high-speed cameras and infrared lightings of different wavelengths, and can achieve fast and almost calibration-free due to the cross-ratio algorithm. We have experimentally validated the eye tracking speed and accuracy, system latency, and demonstrated gaze-oriented projection.

Index Terms: Computing methodologies—Tracking; Hardware—Displays and imagers; Computing methodologies—Mixed / augmented reality; Human-centered computing—Displays and imagers

1 INTRODUCTION

In order to achieve wide field of view and high resolution with low computational costs of display, various researches on gaze-oriented display systems have been attempted. One of the common approaches is directional display; moving a display area according to the user's gaze point [5]. Combining high-resolution display of the gaze point with low-resolution display of peripheral wide area can reduce peripheral information without changing the subjective visual performance. In order to realize this system, it is necessary to achieve high-speed and wide-area in both gaze point estimation and projection techniques. In particular, processing speed is important in both the gaze point estimation and dynamic image projection, because a misalignment between the user's gazing point and the display position may impair the sense of immersion. Based on previous studies on the relationship between latency and perception

in image presentation [3], the latency of the entire projection system should be at most 24.3 ms.

In this paper, we propose a new displaying system that integrates two high-speed methods: gaze point estimation and tracking projection (Fig. 1(a)). For the gaze point estimation method, we apply the almost calibration-free algorithm using corneal reflections [7, 8] and upgrade to high-speed image processing at 500 fps. As the image projection method, we use a high-speed directional projection at 500 fps [5] for gaze-oriented projection.

2 PROPOSED SYSTEM

The overall configuration of the proposed system is shown in Fig. 1(a). High-speed cameras capture the eye and estimate the gaze point, and control projected images around the gaze point. The tracking projection is performed at a speed of 500 fps using a high-speed projector equipped with position control using a galvanometer mirror, as in the method by Nitta [5]. For gaze point estimation, we use dual infrared imaging. The illuminations at the four corners of the screen and coaxial to the camera emit infrared light at different wavelengths, and the synchronized two cameras observe different view (dark and bright pupil) using a beam splitter at 500 fps.

2.1 Gaze point estimation algorithm

The flow of the gaze point estimation is shown in Fig. 2. Specifically, we take the following four steps. (1) We calculate a homography matrix between the two cameras using ChArUco [1]. The homography matrix is used to match the coordinate systems of the feature points in the two camera images. (2) Image processing of bright pupil view; a pupil detection using ellipse fitting and a corneal reflection detection. (3) Image processing of dark pupil view; four corneal reflections detection. (4) Estimate the gaze point on the screen using the information on the pupil and corneal reflection positions obtained (2) and (3).

We use a cross-ratio-based gaze estimation algorithm [8] in the step (4). We utilize the wavelength-separated images in the step (2) and (3) to avoid time-division strategy, and homography-based cross-ratio calculation in the step (4) for more stable estimation. These high-speed image processing assume small target motion from a previous image, and can apply self-window method [2] for even faster and more robust tracking.

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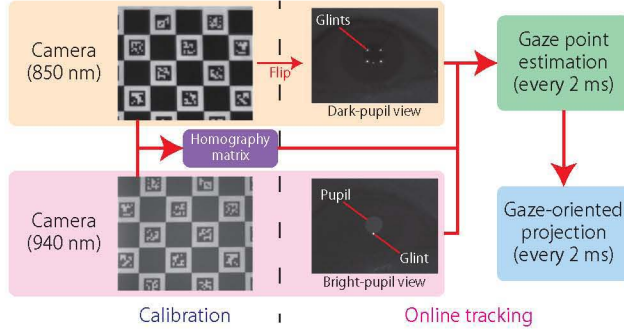


Figure 2: Flow of calibration, the gaze point estimation, and gaze-oriented projection.

Table 1: Estimation Results."B(T)-R(L)" mean that the user looked at positions on the bottom(top)-right(left) corner and center of the screen.

Face	Gaze	Time[ms $\times 10^{-2}$]	X[mm]			Y[mm]	
		Mean(s.d.)	s.d.	RMSE	s.d.	RMSE	
F	C	34.6(10.1)	5.18	11.1	5.61	10.6	
	L-T	48.9(16.0)	7.69	15.5	3.24	33.6	
	R-T	40.7(13.6)	3.87	21.6	6.03	33.6	
	L-B	35.0(10.8)	7.38	30.7	10.78	22.5	
	R-B	34.2(10.7)	10.15	10.3	5.00	12.6	
L	C	38.7(11.2)	4.41	25.1	3.31	3.9	
	L-T	37.4(10.7)	94.20	105.7	62.72	92.0	
	R-T	42.6(12.7)	19.06	21.7	19.85	83.0	
	L-B	40.6(12.4)	6.45	10.0	12.88	23.4	
	R-B	46.3(15.8)	6.74	15.9	6.52	19.6	

3 EXPERIMENT

Two high-speed cameras (500 fps, 592×500 px, monochrome, the focal length was 25 mm) were used for dual infrared imaging. The dichroic mirror for the high-speed camera was longpass (cut-on wavelength, 900 nm). An infrared spotlight (wavelength, 940 nm), was set on the same optical axis of the high-speed cameras using an infrared beam splitter (50R/50T). On the screen side, four IR ring lights (wavelength, 850 nm) were used. The two high-speed cameras were synchronized via trigger I/O. We used the same settings of the tracking projection [5]. A DELL Precision Tower 7910 (Windows 10, Intel Xeon CPU 2 processor 2.40 GHz, RAM 32 GB) was used for system control. The projector was installed at a distance of 2 m from the screen (width, 1030 mm; height, 930 mm). The user sat at a distance of 1.6 m from the eyes to the screen and 0.3 m from the eyes to the camera. The parameter of the gaze point estimation is set to $\alpha = 1.2$ [8] based on the results of the experiment.

Off-line analysis of the gaze point estimation shows that the proposed method achieves sufficient speed and accuracy (Tab. 1). The user gazed at the center of the screen and four other points (right-up, right-down, left-up, and left-down) from two locations: the front ("F") of the screen and a location 20 cm to the left ("L"). The calculation time was less than 1ms in all cases, achieving sufficient speed for the proposed system. The RMSE was about 20 mm, it was worse than the previous research [8] because of the differ of the process of decision of parameters and condition of the experiment. However, this is enough to cover to the effective field of view when used in projection system. The largest error occurred when the user's face was on the left and the gaze point was on the lower right. Because the corneal reflection of the illumination in the corner of the screen was not clear, the reflection of the sclera was mistakenly detected as a corneal reflection.

We have also validated the system latency from eye tracking to

projection using a synchronized stereo camera (500 fps). Moving the gaze and the projected circle from left to right, the average of the difference in time passing through the center was 11.2 ms with a standard deviation of 0.912 ms (5 trials). Finally, we demonstrated the system for a user application (Fig.1(b)). The proposed system works stably even when the user grasps the camera for gaze point estimation, causing hand shake and body movement.

4 DISCUSSION & CONCLUSION

In conclusion, we have proposed a high-speed, wide-area, high-resolution projection system and achieved quick handling of human eye movements. Experiments show that the latency reaches the desired value determined from the previous study [3]. Furthermore, there is a margin of view angle of the projector, the response performance of the system is considered to be sufficient. The dominant factors of the latency of the system are the 3 ms latency of the projector included in the high-speed synchronous projection method [5] and the operation period of the system 2 ms.

However, the area of current system is limited by focus range of the camera and the limitation of the algorithm of capturing the reflection on the surface of the eye. A solution to more wider-area usage is to introduce a mirror-based tracking system [4]. It can be introduced because of the high-speed and the calibration-free nature of the proposed method. Another solution is reducing the directivity of infrared illumination on the screen.

Furthermore, the proposed system is fast enough that it can detect the initial stage of blinking and add exception processing. This can make the system even more practical. Changing the image sensor to a multi-NIR spectral image sensor [6] is also being considered. This will make it possible to capture multiple images with a single camera, which is expected to reduce the size of the camera system, simplify the algorithm, and increase the accuracy.

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